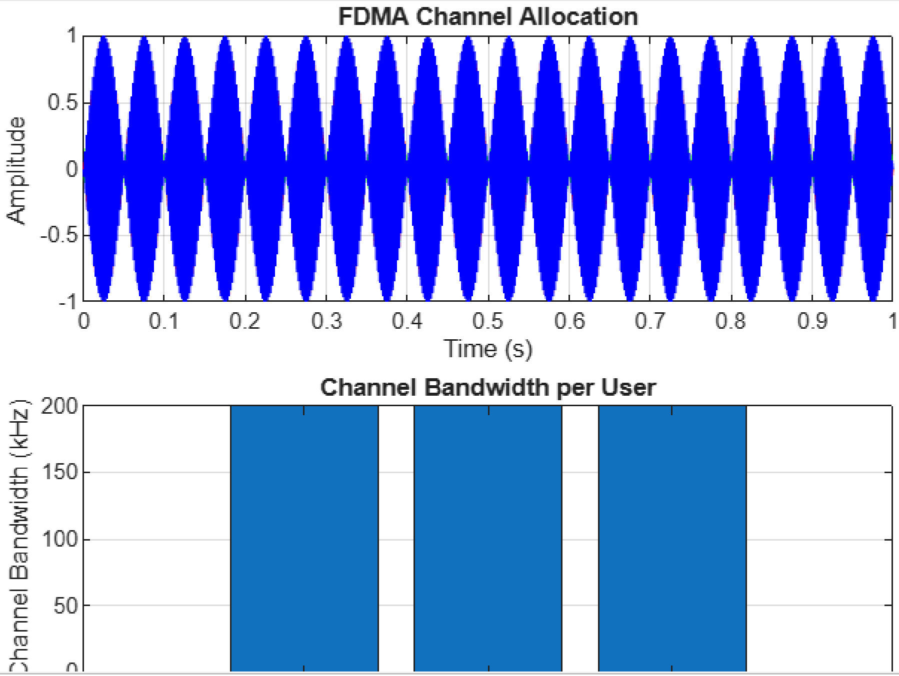
Ex.No:1: Simulation of FDMA in MATLAB

Objective 1:

```
clc;
clear;
close all;
fs = 1000;
t = 0:1/fs:1;
TBW = 800;
Users = 4;
CBW = TBW/Users;
fc = [100 300 500 700];
m = sin(2*pi*10*t);
s1 = m.*cos(2*pi*fc(1)*t);
s2 = m.*cos(2*pi*fc(2)*t);
s3 = m.*cos(2*pi*fc(3)*t);
s4 = m.*cos(2*pi*fc(4)*t);
subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b',t,s4,'m'), grid on
title('FDMA Channel Allocation')
xlabel('Time (s)')
ylabel('Amplitude')
subplot(2,1,2)
bar(1:Users,CBW*ones(1,Users)), grid on
```

```
xlabel('Users')
ylabel('Channel Bandwidth (kHz)')
title('Channel Bandwidth per User')
```

OUTPUT

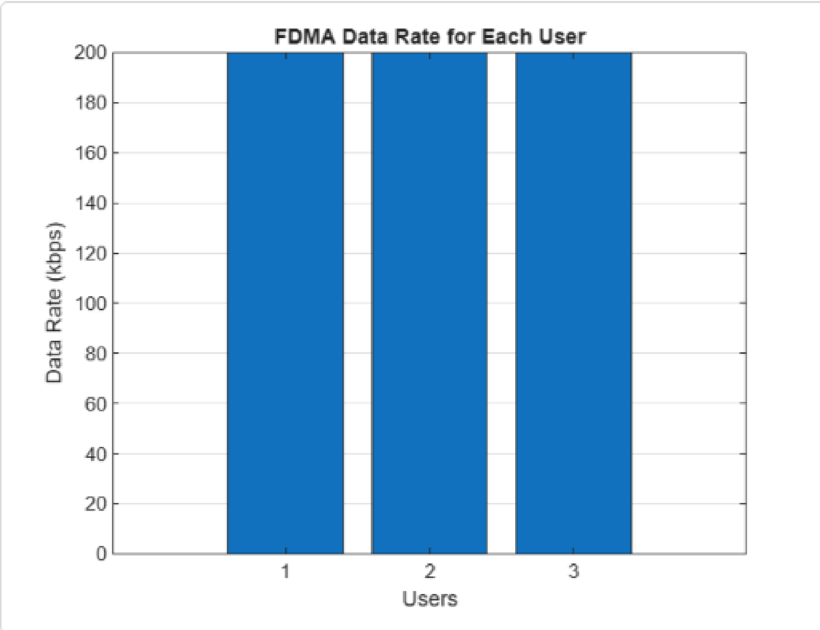


OBJECTIVE 2

```
clc;
clear;
close all;
BW = 200;          % Channel Bandwidth (kHz)
Users = 4;
M = 2;             % BPSK Modulation
DataRate = BW*log2(M);    % Data Rate (kbps)
```

```
Rate = DataRate*ones(1,Users);
disp(' User Data Rate (kbps)')
disp([(1:Users)' Rate'])
figure
bar(1:Users,Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```

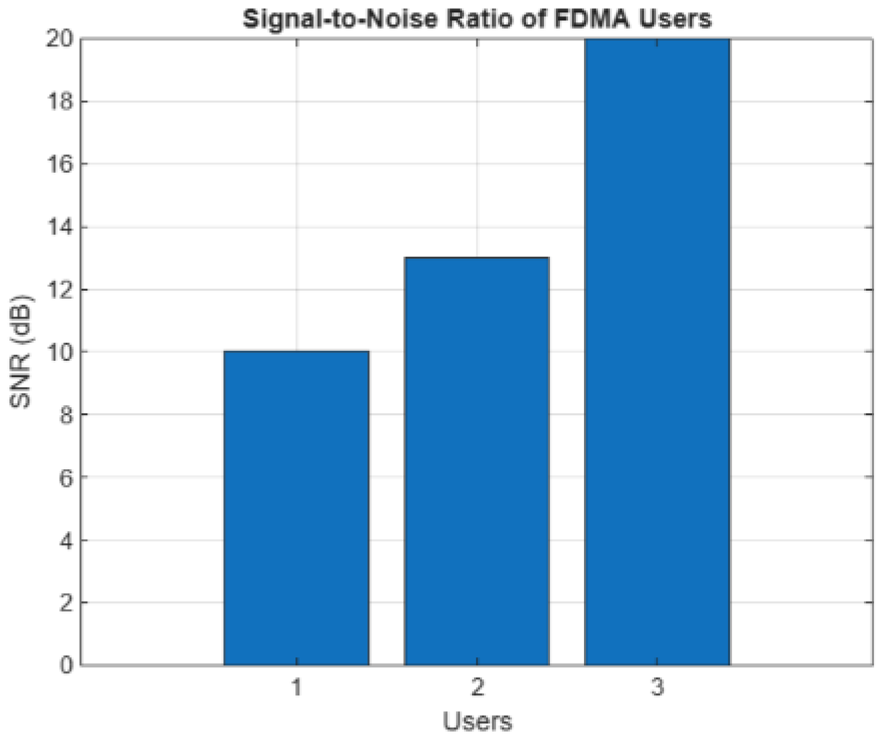
OUTPUT



OBJECTIVE 3

```
clc;
clear;
close all;
Ps = 1;                         % Signal Power (W)
Pn = [0.1 0.05 0.02 0.01];      % Noise Power (W)
Users = 1:4;
SNR = 10*log10(Ps./Pn);         % SNR in dB
disp('User   Noise Power(W)   SNR(dB)')
disp([Users' Pn' SNR'])
figure
bar(Users,SNR)
xlabel('Users')
ylabel('SNR (dB)')
title('Signal-to-Noise Ratio of FDMA Users')
grid on
```

OUTPUT



Signal-to-Noise Ratio of FDMA Users
20
18
16
14
12
10
8
6
4
2
0
SNR (dB)
1
2
3
Users